

UniMate AI

AI-Powered Smart University Platform

Project Documentation

Document No.: 04

Software Requirements Specification (SRS)

Part I

Project Introduction & Overall Description

IEEE 29148 Inspired

Version: 1.0

Prepared By

Abdullah Shabat

Graduation Project

Faculty of Information Technology

Department of Distributed Computing & Smart Device Applications

Islamic University of Gaza

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Document Information

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Version History

Version	Date	Description
1.0	To Be Added	Initial Version

Table of Contents

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1. Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of the UniMate AI platform. This document serves as the primary reference for developers, testers, supervisors, and future contributors throughout the software development lifecycle.

The specification aims to ensure that all stakeholders share a common understanding of the system before implementation begins.

2. Scope

UniMate AI is an Artificial Intelligence-powered academic platform designed to simplify university education through a unified ecosystem consisting of:

- Android Mobile Application
- ASP.NET Core REST API
- React Web Dashboard
- PostgreSQL Database
- AI Assistant
- Cloud Storage

The system supports students, instructors, and administrators by integrating academic management with intelligent educational services.

3. Definitions

Term	Description
AI	Artificial Intelligence
API	Application Programming Interface
JWT	JSON Web Token
REST	Representational State Transfer
RBAC	Role-Based Access Control
MVP	Minimum Viable Product
Dashboard	Administrative Web Application
Backend	ASP.NET Core REST API
Client	Android Application
Storage	Cloud File Storage

4. References

This document is based on:

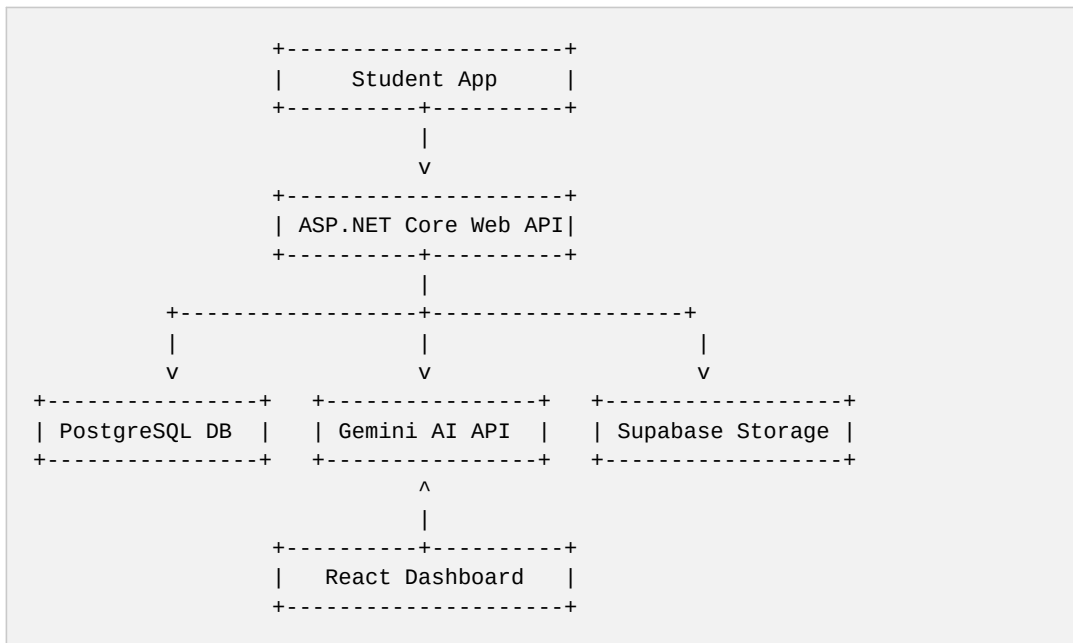
- IEEE 29148 Software Requirements
- REST API Design Principles
- OWASP Security Guidelines
- Clean Architecture Principles
- Android Development Guidelines
- ASP.NET Core Documentation

5. Product Perspective

UniMate AI is designed as a distributed system composed of independent but integrated components.

Each component has a clearly defined responsibility, allowing the platform to scale and evolve without major architectural changes.

6. System Context



7. Product Functions

Version 1.0 provides the following high-level capabilities.

Authentication

- User Registration
- Login
- Logout
- Password Reset
- Email Verification

User Management

- Student Profiles
- Doctor Profiles
- Administrator Profiles

Academic Management

- Courses
- Lecture Materials
- Assignments
- Course Files

Artificial Intelligence

- AI Chat
- Lecture Summarization
- Academic Assistance

File Management

- Upload Files
- Download Files
- Secure Cloud Storage

Administration

- User Management
- Department Management
- Reports
- Dashboard Statistics

8. User Classes

Student

Responsibilities:

- Access courses
- Submit assignments
- Ask AI questions
- Manage files
- View notifications

Doctor

Responsibilities:

- Create courses
- Upload lectures
- Manage assignments
- Review student submissions

Administrator

Responsibilities:

- Manage users
- Manage departments
- Monitor reports
- Configure system settings

9. Operating Environment

Component	Environment
Mobile App	Android 10+
Dashboard	Modern Web Browser
Backend	Linux Server / Docker
Database	PostgreSQL
Storage	Supabase
AI	Gemini API

10. Design Constraints

The project must comply with the following constraints:

- Clean Architecture.
- MVVM for Android.
- RESTful APIs.
- JWT Authentication.
- Firebase Authentication.
- PostgreSQL Database.
- Version Control using Git.
- Incremental Development Strategy.

11. Assumptions

The following assumptions apply:

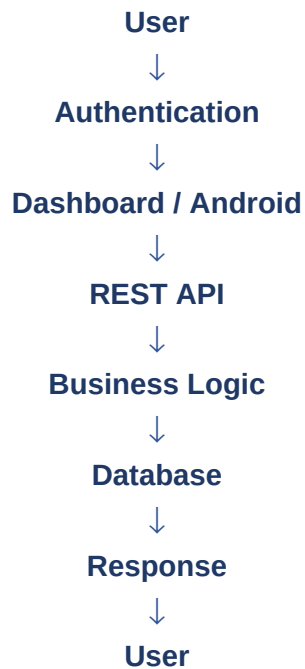
- Internet connectivity is available.
- Firebase services remain operational.
- Gemini API quota is sufficient.
- Supabase services remain accessible.
- Users possess basic smartphone skills.

12. Dependencies

The project depends on:

- Firebase Authentication
- Gemini API
- Supabase Storage
- PostgreSQL
- Android SDK
- ASP.NET Core Runtime

13. High-Level System Workflow



14. System Modules Overview

Module	Description
Authentication	Secure user authentication and authorization
User Management	Profile and account management
Course Management	Academic course administration
Assignment Management	Assignment creation and submission
AI Module	Intelligent educational assistant
File Management	Cloud-based document management
Notification Module	User notifications and alerts
Dashboard Module	Administrative web interface

15. Version 1.0 Deliverables

The first release includes:

- Android Application
- Backend API
- Dashboard
- Authentication
- AI Chat
- Course Management
- Assignment Management
- File Storage
- User Profiles

16. Excluded Features

The following features are planned for future versions:

- AI Quiz Generator
- Flashcard Generator
- Attendance Tracking
- Calendar Synchronization
- Video Conferencing
- University ERP Integration
- Multi-language Support

17. Success Criteria

The system shall be considered ready when:

- Authentication operates correctly.
- APIs pass testing.
- Mobile application communicates with backend.
- Dashboard communicates with backend.
- AI module functions correctly.
- File management operates securely.
- Demonstration scenarios execute successfully.

18. Conclusion

This document establishes the overall software requirements for UniMate AI and serves as the foundation for all subsequent technical specifications. By clearly defining the system context, stakeholders, scope, constraints, and high-level functionality, it provides a structured basis for architecture, implementation, testing, and deployment.

Design Decisions

Decision	Justification
Clean Architecture	Promotes maintainability, scalability, and separation of concerns.
MVVM	Encourages a clear separation between UI and business logic in Android.
ASP.NET Core	Provides a robust framework for building secure and high-performance APIs.
React	Enables a responsive and component-based administrative dashboard.
PostgreSQL	Offers strong relational capabilities and long-term scalability.
Firebase Authentication	Simplifies secure user authentication and account management.
Supabase Storage	Provides secure cloud file storage with controlled access.
Gemini API	Delivers AI capabilities suitable for educational assistance.

End of Document

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